



ASSESSMENT - I

Project Proposal Report

Project Title:

Intelligent Cognitive Health Monitoring Through
Ambient IoT and Immersive Visualization

PREPARED BY: HESSA KAMAL

STUDENT ID: 32146983

SUPERVISOR: DR. MOHAMED SHAMEEM

COURSE: COMPUTER SCIENCE

TABLE OF CONTENTS

Introduction	3
Aim & Objectives	3
Literature Review	
Background	4
Research Gap	6
Conclusion	
Critical Evaluation	7
Justification	7
Project Outline	
Tools & Technologies	8
Risks	8
Project Timeline	9
Appendix	
Abbreviations	10
Project Progress Form I	11
References	13

Introduction

Mental health challenges among students and professionals have intensified, with studies showing depression, anxiety, and stress affect approximately 40% of students during their academic careers (*Beiter et al., 2021*). The COVID-19 pandemic exacerbated these issues.

Huckins et al. (2020) documented significant deteriorations in student mental health during early pandemic phases through smartphone monitoring, revealing that early detection systems could identify concerning patterns before clinical intervention became necessary. Traditional stress management relies on self-awareness and self-reporting, which are inherently limited. People often fail to recognize deteriorating mental states until physical symptoms emerge or performance declines. By the time stress becomes consciously apparent, intervention becomes more difficult and recovery takes longer. Recent advances in ML and behavioral analytics present opportunities to transform mental health monitoring with proactive approaches (*Thakkar et al., 2024*).

Aim

Design, develop, and evaluate an accessible cognitive health monitoring system using behavioral analytics (keyboard and mouse dynamics) and machine learning to detect stress patterns and provide personalized, privacy-preserving ambient feedback

Objectives

1. Design and implement an **ML model** capturing keyboard dynamics and mouse dynamics on Raspberry Pi.
2. Develop and train **LSTM autoencoder** on public keystroke dynamics datasets, achieving reconstruction accuracy $\geq 85\%$, then fine-tune using personal baseline data.
3. **Ambient Feedback:** Visualize data using 3D house morphology reflects mental state and ambient lamp shows cognitive load spectrum.
4. **Evaluate the system** with tests on user data privacy, feedback mechanism, and ML model.



Literature Review



Driven by the escalating mental health crisis and advances in passive sensing technologies, understanding everyday digital behaviors to reveal cognitive states have brought multiple disciplines together for the shared goal of developing accessible, effective wellness technologies.

Research Question: Can ML models accurately detect cognitive stress from minimal behavioral data alone, to provide daily monitoring without specialized physiological sensors/environment?

Background

The potential for early detection through behavioral pattern analysis has driven essential studies into ML approaches for mental health prediction.

Chung and Teo (2022) provided a comprehensive taxonomy of ML applications for mental health, systematically reviewing studies to categorize approaches by data modality, algorithmic techniques, and application domains. Their analysis revealed that while physiological sensor-based approaches achieved the highest accuracies (85-92%), behavioral analytics approaches using digital patterns achieved competitive performance (72-85%) while offering somewhat better accessibility and lower user burden.

The review by Khoo et al. (2024) examined multimodal mental health detection using passive sensing approaches, analyzing 47 studies published between 2019-2024. Their comprehensive analysis revealed that behavioral data from daily digital activities, i.e., typing patterns, mouse movement dynamics, application usage, and digital communication patterns, achieved detection accuracies comparable to physiological sensors when multiple behavioral modalities were combined. Single-modality approaches (keyboard-only or mouse-only) typically achieved 65-75% accuracy, but multimodal combining keyboard, and mouse reached 80-87% accuracy. Impressively, their review also documented that edge processing significantly increased user acceptance rates compared to cloud-based approaches. They concluded that the field was advancing to “invisible” systems that derive insights from natural behaviors without requiring users to wear or charge devices. This is a vision that aligns closely with this project's ambient computing motive.

Nath et al. (2020) studied ML-based real-time stress monitoring solutions across multiple contexts – mobile apps, and computer systems. Their comparison revealed critical tradeoffs: smartphone-based monitoring achieved moderate accuracy (75-83%); computer-based monitoring achieved comparable accuracy (72-84%). For students spending 6-12 hours daily at computers, this second approach offered practicality. They emphasized that successful development used edge computing by processing data locally rather than transmitting to cloud servers, both to reduce latency and to address privacy concerns among users.

Yang and Qin (2021) conducted an extensive review of emotion recognition methods from keystroke, mouse, and touchscreen dynamics, synthesizing findings from **67** studies spanning varying contexts. Their analysis showed behavioural signals associated with different emotional states: stressed typing exhibited increased inter-key interval variation, increased error rates, and unstable typing rhythm. For mouse dynamics, stress manifested as increased movement jerkiness, more frequent direction changes, and longer decision pauses before clicks. Critically, they found that these patterns were largely unconscious and participants couldn't suppress stress-induced behavioral changes, making these signals more authentic than self-reports sensitive to social bias.

Maalej et al. (2022) researched keystroke dynamics for real-time emotion recognition, collecting data from 45 participants across different emotional states. Their findings validated that stressed typing exhibited characteristic signals: mean inter-key interval increased – people typed slower under stress, but standard deviation also increased 25-35%, creating an identifiable pattern of hesitant, irregular typing. These patterns emerged within 2-3 minutes of stress onset and persisted for 5-15 minutes after stressor removal, providing a window for detection and intervention.

Machado-Jaimes et al. (2022) developed an intelligent system for monitoring and diagnosing wellbeing that combined physiological sensors with behavioral analytics for comprehensive assessment. Their multimodal system achieved 86% accuracy in detecting elevated stress states, with behavioral features contributing 40% of the prediction despite being only 30% of total feature, suggesting behavioral signals carried information density comparable to or even exceeding physiological data.

Thakkar et al. (2024) reviewed artificial intelligence applications in positive mental health, analyzing systems not designed to diagnose disorders but to promote wellness. They identified key design principles of successful wellness technologies from clinical tools: these systems prioritized user empowerment over medical expertise, provided actionable guidance rather than assessments, integrated seamlessly into existing routines instead of a change, and emphasized transparency enabling users to understand and verify system operation. They documented that systems with these traits maintained 3-5 times higher engagement rates compared to clinical-feeling health monitoring applications, showing that design philosophy matters as much as technical accuracy for users.



Research Gap

Through my comprehensive review of the literature, I have identified several critical gaps that the proposed system addresses.

Accessibility barrier: The systems achieving the highest detection accuracies are precisely those requiring the most expensive equipment. *This creates an inverse relationship where technology benefits those least in need, who research suggests experience higher stress levels.* My project addresses this gap by demonstrating that meaningful stress detection is achievable using only standard equipments that most students already use, requiring merely a Raspberry Pi kit as additional hardware.

Real-world deployment: Laboratory studies achieve impressive accuracies under controlled conditions with standardized tasks and supervised data collection; *few researchers have documented system performance during actual daily computer usage with diverse tasks, and varying workloads.*

Privacy: While multiple papers discuss privacy theoretically and acknowledge user concerns, *most implemented systems described require cloud connectivity for core functionality.* This approach fundamentally contradicts users' desire for verifiable control over behavioral data.

Ambient feedback: The majority of stress detection systems reviewed provide feedback through notifications or dashboard visualizations requiring active checking. This distracting design creates two problems: *alert fatigue where users begin ignoring notifications, and paradoxical stress where monitoring becomes a source of anxiety.* My ambient LED feedback, providing awareness through color shifts without demanding attention, represents a very different design that I have not observed in existing behavioral stress monitoring computer systems.

Conclusion

Critical Evaluation

I identified some critical challenges, including limiting practical deployment, data privacy concerns, and the need for model interpretability so users can understand and trust these assessments. The literature emphasized that addressing the accessibility challenge by developing systems that work with common equipments rather than specialized tools – represented an essential research priority.

Justification

Having conducted comprehensive literature review and designed the system architecture, I believe my system represents a feasible and valuable contribution to accessible cognitive wellness monitoring. Multiple independent research streams validate that stress manifests in measurable behavioral patterns (*Maalej et al., 2022; Yang and Qin, 2021*), LSTM autoencoders detect anomalies in time-series data (*Khoo et al., 2024*), and processing on Raspberry Pi is capable of real-time ML inference (*Nath et al., 2020*). These techniques can be integrated into a system addressing identified gaps.

Successfully completing this project will demonstrate advanced skills valued in the industry. The 4-month timeline is ambitious but adjustable if I maintain disciplined progress and seek guidance when encountering obstacles.

I chose behavioral analytics over physiological sensors because of the accessibility implications. I want to demonstrate that meaningful support is possible using ubiquitous equipment. The edge computing architecture aligns with my values about technology and privacy. I believe in control over intimate personal data, rather than trusting corporate promises. Finally, I chose ambient feedback since wellness technology should reduce cognitive burden, not add to it.

I am realistic about what this project can achieve – it will not solve the mental health crisis, and it will not produce a commercial-ready product. But if I successfully demonstrate that inexpensive hardware can provide meaningful stress awareness, and enable others to use this work, then I will have made a worthy contribution to making cognitive wellness more accessible.

Project Outline



Tools & Technologies

Hardware: Raspberry Pi 4 Model B, LED components, standard computer with keyboard/mouse

Software: Python, TensorFlow, pynput, NumPy, Pandas, Three.js / Unity

Datasets: CMU Keystroke Dynamics Benchmark, Balabit Mouse Dynamics Dataset

Development: VS Code, Git, Jupyter

Risks

These are the primary risks identified and mapped into a table.

Risk	Probability	Impact	Mitigation
LSTM training difficulties	High	High	Start with simpler models, early supervisor consultation
Raspberry Pi performance issues	High	Medium	Early testing, optimization, model compression
Setup technical barriers	Medium	Low	Hands-on assistance, excellent documentation
Low detection accuracy	Medium	Medium	Set realistic expectations, analyze failure modes

Table 1. Risk Analysis Chart

Project Timeline

The project has been divided into manageable tasks spanning a 4-month timeline. It is represented using the Gantt Chart below.

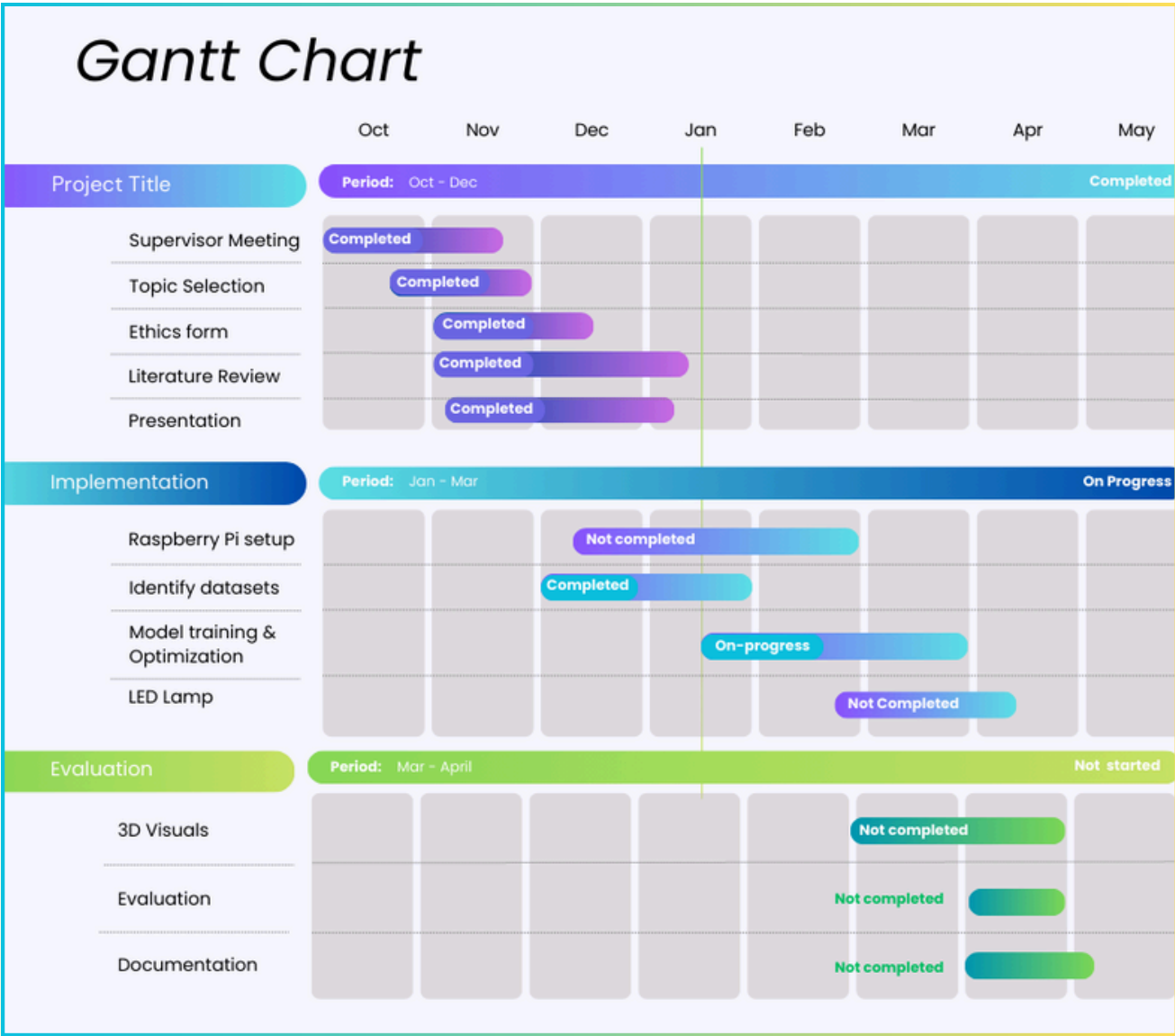
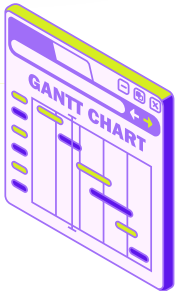


Fig 1. Gantt Chart

References

Khoo, L.S., Lim, M.K., Chong, C.Y. and McNaney, R. (2024). Machine Learning for Multimodal Mental Health Detection: A Systematic Review of Passive Sensing Approaches. *Sensors*, [online] 24(2), p.348. doi:<https://doi.org/10.3390/s24020348>.

Thakkar, A., Gupta, A. and De Sousa, A. (2024). Artificial intelligence in positive mental health: a narrative review. *Frontiers in digital health*, [online] 6(1280235). doi:<https://doi.org/10.3389/fdgth.2024.1280235>.

Chung, J. and Teo, J. (2022). Mental Health Prediction Using Machine Learning: Taxonomy, Applications, and Challenges. *Applied Computational Intelligence and Soft Computing*, [online] 2022, p.e9970363. doi:<https://doi.org/10.1155/2022/9970363>.

Huckins, J.F., DaSilva, A.W., Wang, W., Hedlund, E., Rogers, C., Nepal, S.K., Wu, J., Obuchi, M., Murphy, E.I., Meyer, M.L., Wagner, D.D., Holtzheimer, P.E. and Campbell, A.T. (2020). Mental Health and Behavior During the Early Phases of the COVID-19 Pandemic: A Longitudinal Mobile Smartphone and Ecological Momentary Assessment Study in College Students (Preprint). *Journal of Medical Internet Research*, [online] 22(6). doi:<https://doi.org/10.2196/20185>.

Maalej, A., Kallel, I. and Sanchez Medina, J.J. (2022). Investigating Keystroke Dynamics and Their Relevance for Real-Time Emotion Recognition. *SSRN Electronic Journal*. doi:<https://doi.org/10.2139/ssrn.4250964>.

Machado-Jaimes, L.-G., Bustamante-Bello, M.R., Argüelles-Cruz, A.-J. and Alfaro-Ponce, M. (2022). Development of an Intelligent System for the Monitoring and Diagnosis of the Well-Being. *Sensors*, 22(24), p.9719. doi:<https://doi.org/10.3390/s22249719>.

Yang, L. and Qin, S.-F. (2021). A Review of Emotion Recognition Methods From Keystroke, Mouse, and Touchscreen Dynamics. *IEEE Access*, 9, pp.162197–162213. doi:<https://doi.org/10.1109/access.2021.3132233>.

Nath, R.K., Thapliyal, H., Caban-Holt, A. and Mohanty, S.P. (2020). Machine Learning Based Solutions for Real-Time Stress Monitoring. *IEEE Consumer Electronics Magazine*, [online] 9(5), pp.34–41. doi:<https://doi.org/10.1109/MCE.2020.2993427>.

Beiter, R., Nash, R., McCrady, M., Rhoades, D., Linscomb, M., Clarahan, M. and Sammut, S. (2015). The prevalence and correlates of depression, anxiety, and stress in a sample of college students. *Journal of Affective Disorders*, [online] 173(1), pp.90–96. doi:<https://doi.org/10.1016/j.jad.2014.10.054>.

Appendix

Abbreviations

ML - Machine Learning

LSTM - Long short-term memory

LED - Light Emitting Diode

DL - Deep Learning

Project Progress Form I

University of West London

School of Computing and Engineering

Project Progress Form 1



Module code: **CP6UA460**

Student Number: 32146983	Submission Date: 06 / 01 / 2026
Student's Surname: -	Student's Forename: HESSA KAMAL
Project Supervisor's Name: Dr. MOHAMED SHAMEEM	

Please note: Work presented for the project must be the student's own. Plagiarism is where a student copies work from another source, published or unpublished (including the work of a fellow student) and fails to acknowledge the influence of another's work or to attribute quotes to the author. Plagiarism is an academic offence.

I confirm this is my own work

Student's signature

Project title

Intelligent Cognitive Health Monitoring Through Ambient IoT and Immersive Visualization

Progress made since project was approved

On receiving approval, I promptly completed a literature review to identify research gaps, relevant methodologies, and create a project plan. I completed the ethics and approval forms, as well as prepared and presented the literature review. Currently, I have chosen algorithms, identified the datasets, and outlined the methodology. Additionally, I've already started working on creating models.

List of meeting dates with supervisor (to be completed by student)

- | | | | |
|--------------------|------------------|--|--|
| Meeting I | (19 / 11 / 2025) | | [On-Campus: Meeting with the Supervisor] |
| Meeting II | (03 / 12 / 2025) | | [On-Campus: Project Outline Submission] |
| Meeting III | (09 / 12 / 2025) | | [On-Campus: Project Topic Discussion] |
| Meeting IV | (16 / 12 / 2025) | | [On-Campus: Project Title Confirmation] |
| Meeting V | (23 / 12 / 2025) | | [On-Campus: Presentation Slides Review] |

Fig 2.1 Progress Form 1

Comments and further actions (to be completed by supervisor before submission)

- Regular conversations between the student and the supervisor were held.
- Literature review was compiled and the draft proposal was submitted.
- The progress is on schedule

	Signatures	Date
Student		06 / 01 / 2026
Supervisor (before submission)		06 / 01 / 2026

Assessment (Project Progress Form) Marking criteria:

Criteria	Issues	Mark
Project Progress Form 1	Form 1: The form submitted on time and signed by the supervisor. There is sufficient evidence provided to show the project has been progressed with Literature Review and the student has attended regular meetings with the supervisor. At least 3 meetings reported.	10

Fig 2.2. Progress Form 1